

PMGI SF5 / S1813 Bilayer Lift-Off Protocol

With DMSO Lift-Off for EGOFET Electrode Patterning

e-MolMat Group – ICMAB-CSIC
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Abstract

This document describes an improved lift-off protocol for patterning Cr/Au source-drain and gate electrodes on Kapton substrates. The method replaces the standard single-layer Shipley S1813 process with a **PMGI SF5 / S1813 bilayer** stack, combined with a **DMSO-based lift-off** replacing the acetone/ultrasonic process. The bilayer creates a controlled undercut profile during development that prevents metal bridging on resist sidewalls, eliminating the need for manual cotton-swab removal of residual gold from transistor channels.

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1 Introduction

1.1 Problem Statement

The current EGOFET fabrication protocol uses a single-layer Shipley S1813 positive photoresist for lift-off patterning of Cr/Au electrodes. This produces vertical or slightly positive sidewall profiles, causing the evaporated metal to form a continuous film on the resist sidewalls. During lift-off, the acetone/ultrasonic baths cannot fully dissolve the resist underneath this metal “fence,” leaving gold residue in the Source–Drain channels. This residue must be removed manually with acetone-soaked cotton swabs, which frequently damages the electrode pattern.

1.2 Proposed Solution

A **bilayer resist stack** consisting of:

- **PMGI SF5** (bottom layer): A polydimethylglutarimide-based lift-off resist that dissolves isotropically in alkaline developer, creating a controlled undercut beneath the imaging resist.
- **Shipley S1813** (top layer): The standard positive imaging resist, processed identically to the current protocol.

The undercut profile ensures that the evaporated metal film is **discontinuous** at the resist edges, allowing the solvent to penetrate and lift off the unwanted metal cleanly. DMSO at 60 °C replaces acetone/ultrasonic for a gentler, more effective strip.

1.3 Required Materials

Table 1: Materials and chemicals required.

Material	Specification	Status
PMGI SF5	MicroChem lift-off resist	In stock
Shipley S1813	Positive photoresist	In stock
MF-319	TMAH developer (2.2%)	In stock
DMSO	Dimethyl sulfoxide, ULSI grade	To purchase
IPA	Isopropanol, cleanroom grade	In stock
DI water	18.2 M Ω · cm	Available
Kapton HN	75 μ m polyimide foil	In stock
Si carrier wafer	4-inch, for mounting	Available

2 Process Overview

The bilayer lift-off process consists of the following major steps:

Step 1: Substrate preparation and mounting

Step 2: PMGI SF5 coating and prebake

Step 3: S1813 coating and soft bake

Step 4: Exposure (Microwriter ML3)

Step 5: Development (single-step, MF-319)

Step 6: Cr/Au evaporation

Step 7: DMSO lift-off

Step 8: Post-lift-off cleaning

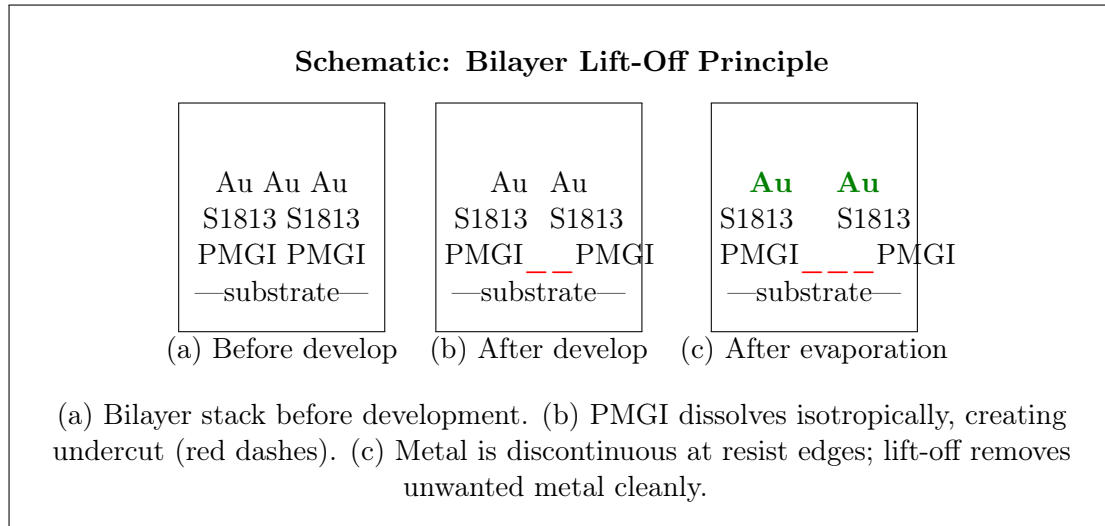


Figure 1: Bilayer lift-off process schematic.

3 Detailed Process Steps

3.1 Substrate Preparation and Mounting

- a) Cut Kapton foil to $7\text{ cm} \times 7\text{ cm}$ pieces. Allow cut pieces to flatten on a clean surface for 2–3 days before use to avoid ridges.
- b) Clean the Kapton substrate:
 - i) Rinse with acetone
 - ii) Rinse with isopropanol
 - iii) Dry thoroughly with N_2 gun (front and back)
- c) Mount the Kapton on a clean Si carrier wafer using Kapton adhesive tape strips at the edges. Leave **gaps** between strips so air can escape during evaporation.
- d) Ensure both Kapton and carrier are **completely dry** — trapped humidity at the interface ruins exposure and evaporation.

Important

Use **two different Si carriers**: one for photolithography (coating + exposure + development) and a fresh one for evaporation. This prevents adhesive residue from contaminating the evaporator.

3.2 PMGI SF5 Coating and Prebake

PMGI SF5 Parameters

Parameter	Value
Spin speed	3000
Spin time	45 s
Acceleration	4000 s^{-1}
Expected thickness	$0.5 \mu\text{m}$ to $0.7 \mu\text{m}$
Prebake temperature	170°C
Prebake time	5 min
Bake mode	Contact hotplate

- Clean the spin coater chuck and support with acetone.
- Place a square of double-sided tape under the carrier wafer to prevent movement during spinning.
- Apply PMGI SF5 by pouring directly from the bottle (avoid pipette to prevent bubbles). Cover the wafer surface completely.
- Press START immediately after dispensing.
- Transfer to hotplate for prebake at 170°C for 5 min.

Tip: Prebake Temperature

The prebake temperature is the **primary control** for undercut rate. At 170°C with MF-319, the undercut rate is approximately $33 \text{ \AA s}^{-1}$ (0.33 nm s^{-1}). This provides a good balance between process control and development speed. If undercut is too aggressive (features lifting off), increase prebake to 190°C ($28 \text{ \AA s}^{-1}$). If undercut is insufficient (metal residue remains), decrease to 150°C ($42 \text{ \AA s}^{-1}$).

3.3 S1813 Coating and Soft Bake

S1813 Parameters

Parameter	Value
Spin speed	5000
Spin time	25 s
Acceleration	4000 s^{-1}
Expected thickness	$1.3 \mu\text{m}$
Soft bake temperature	75°C
Soft bake time	1 min
Bake mode	Contact hotplate

- Apply S1813 directly on top of the baked PMGI layer. **No intermixing occurs** — PMGI is insoluble in the ethyl lactate solvent of S1813.
- Spin at 5000 for 25 s.
- Soft bake at 75°C for 1 min.

Do Not

Do not press manually on the spinner lid during coating. Do not use HMDS — PMGI has excellent adhesion without primers.

3.4 Exposure

- Expose the pattern using the Microwriter ML3 laser direct-write system.
- Use the **same dose** as the current S1813-only process.
- Recommended focus calibration: 9 points with offsets (0, 0), (± 30 , 0), (0, ± 30), (± 20 , ± 20) μm .

3.5 Development (Single-Step Bilayer)**Development Parameters**

Parameter	Value
Developer	MF-319 (TMAH 2.2% with surfactant)
Mode	Immersion, manual stirring
Time	60 s to 90 s
Rinse	DI water, ≥ 300 mL
Dry	N ₂ gun, front and back

- Carefully remove the Kapton foil from its Si carrier (do not bend).
- Immerse in MF-319 with the patterned face **up**.
- Stir manually for 60 s to 90 s.
- Rinse thoroughly with DI water (300 mL or more).
- Dry **front and back** with N₂ gun — residual water affects evaporation quality.

Understanding the Development

MF-319 develops **both layers simultaneously**:

- The exposed S1813 dissolves in the developer (positive-tone behavior).
- The underlying PMGI dissolves **isotropically** — sideways as well as downward — creating the undercut.

The undercut depth is:

$$d_{\text{undercut}} = r_{\text{PMGI}} \times t_{\text{develop}}$$

where $r_{\text{PMGI}} \approx 33 \text{ \AA s}^{-1}$ at 170 °C prebake with MF-319.

For 60 s: $d \approx 2.0 \mu\text{m}$ of undercut.

For 90 s: $d \approx 3.0 \mu\text{m}$ of undercut.

Both values are sufficient for 45 nm of metal deposition.

Critical

Do **not** over-develop. Excessive undercut can cause small features ($< 10 \mu\text{m}$) to detach from the substrate. Start with 60 s and increase only if metal residue persists after lift-off.

3.6 Cr/Au Evaporation

- a) Re-mount the developed foil on a **fresh** Si carrier (different from the one used for photolithography).
- b) Provide the evaporator technician with the following parameters:

Evaporation Parameters — Request to Technician

Parameter	Value
Adhesion layer	Cr, 5 nm
Electrode layer	Au, 40 nm
Cr deposition rate	$\leq 0.2 \text{ \AA s}^{-1}$
Au deposition rate	$\leq 0.5 \text{ \AA s}^{-1}$
Base pressure	$\leq 5 \times 10^{-6}$
Substrate rotation	None (incidence normal)

Quality Check

After evaporation, the Au film should appear **gold-colored and reflective**. If it looks pinkish, opaque, or matte, inform the technician — this indicates oxidation or contamination and will cause poor SAM formation and device performance.

3.7 DMSO Lift-Off

DMSO Lift-Off Parameters

Parameter	Value
Solvent	DMSO (dimethyl sulfoxide), ULSI grade
Temperature	60 °C
Duration	2 h to 3 h
Ultrasonic	None (passive soak only)
Agitation	Gentle manual swirling every 30 min

- a) Place the sample in a glass beaker with fresh DMSO.
- b) Cover with a watch glass or petri dish (DMSO has low vapor pressure but should still be contained).
- c) Place on hotplate at 60 °C.
- d) Soak for 2 to 3. Swirl gently every 30 min.
- e) The metal “skin” on top of the resist will begin to detach and float to the surface.
- f) Once lift-off is complete, remove sample and rinse immediately with IPA.

Why DMSO, not Acetone?

DMSO is preferred over acetone for the following reasons:

- **No ultrasonic required:** The undercut profile allows DMSO to penetrate under the metal without mechanical agitation, preventing damage to delicate electrode patterns.
- **Better penetration:** DMSO has lower surface tension and higher boiling point than acetone, allowing it to wick into the undercut gap more effectively.
- **Non-toxic:** DMSO is not classified as toxic, unlike NMP (the traditional lift-off solvent).
- **Compatible with Kapton:** DMSO does not attack polyimide substrates.

3.8 Post-Lift-Off Cleaning

- a) Rinse with IPA: 2 baths $\times$ 15 min each, fresh IPA per bath.
- b) Dry with N₂ gun.
- c) Inspect under microscope: verify clean channels with no residual gold.

Expected Result

After successful lift-off, gold should be present **only** in the patterned electrode areas. The Source–Drain channels should be **completely clean** — no gold islands, no metal fences, no need for cotton-swab removal. If residue persists, see Section 4.

4 Troubleshooting

Table 2: Common issues and solutions.

Symptom	Likely Cause	Solution
Gold residue in channels	Insufficient undercut	Increase develop time by 15 s increments. Check PMGI pre-bake temperature.
Electrode pattern de-taching	Excessive undercut	Decrease develop time or increase PMGI prebake temperature to 190 °C.
Lift-off takes >3 h	DMSO too cold or undercut too small	Verify hotplate temperature. Increase develop time.
Pinkish Au film	Evaporator contamination	Re-evaporate. Inform technician.
Poor SAM formation	Au surface contaminated	Perform UV-ozone clean (25 min) immediately before SAM.
PMGI not adhering to Kapton	Surface contamination	Ensure dehydration bake at 200 °C/5 min before PMGI coating.

5 Process Comparison

Table 3: Comparison: Current protocol vs. Bilayer protocol.

Parameter	Current (S1813 only)	Bilayer (PMGI + S1813)
Photoresist layers	1 (S1813)	2 (PMGI SF5 + S1813)
Sidewall profile	Vertical	Undercut (re-entrant)
Lift-off solvent	Acetone + ultrasonido	DMSO at 60 °C
Lift-off time	3 × 15 min + IPA baths	2 h to 3 h passive
Cotton swab needed	Yes (frequently)	No
Pattern damage risk	High	Low
Developer	MF-319	MF-319 (same)
Exposure tool	Microwriter ML3	Microwriter ML3 (same)
Additional equipment	None	Hotplate capable of 170 °C
New chemicals	None	DMSO only

6 References

- [1] MicroChem Corp. *PMGI Resists — Technical Data Sheet*. Kayaku Advanced Materials, 2021. Available: <https://kayakuam.com/product/structure/lift-off-resists>
- [2] MicroChem Corp. *LOR Lift-Off Resists — Technical Data Sheet*. Kayaku Advanced Materials, 2021.

- [3] MicroChemicals GmbH. *DMSO (ULSI) — Product Information*. Available: <https://www.microchemicals.com>
- [4] Nanyte. *Lift-off lithography, step by step*. Tutorial, 2025–2026. Available: <https://nanyte.com/photoresist-tutorials/lift-off>
- [5] Bonafe, F.P. et al. “Dissipative charge transport in organic mixed ionic-electronic conductor channels.” *Nature Communications* **16**, 4537 (2025). 10.1038/s41467-025-57528-9
- [6] Ruiz-Molina, S. PhD Thesis, ICMAB-CSIC, 2024. “EGOFETs for biosensing.” Chapter 3, Section 2.2.
- [7] Ricci, S. PhD Thesis, ICMAB-CSIC, 2020. “Organic electrochemical transistors for biosensing.” Chapter 6.
- [8] Fraunhofer IPMS. “OFET Substrates — Test Chips for Material Characterization.” Protocol for resist removal, 2026.
- [9] University of Minnesota Nano Center. “PMMA/PMGI Bilayer E-Beam Liftoff Process.” Institutional protocol. <https://cse.umn.edu/mnc>